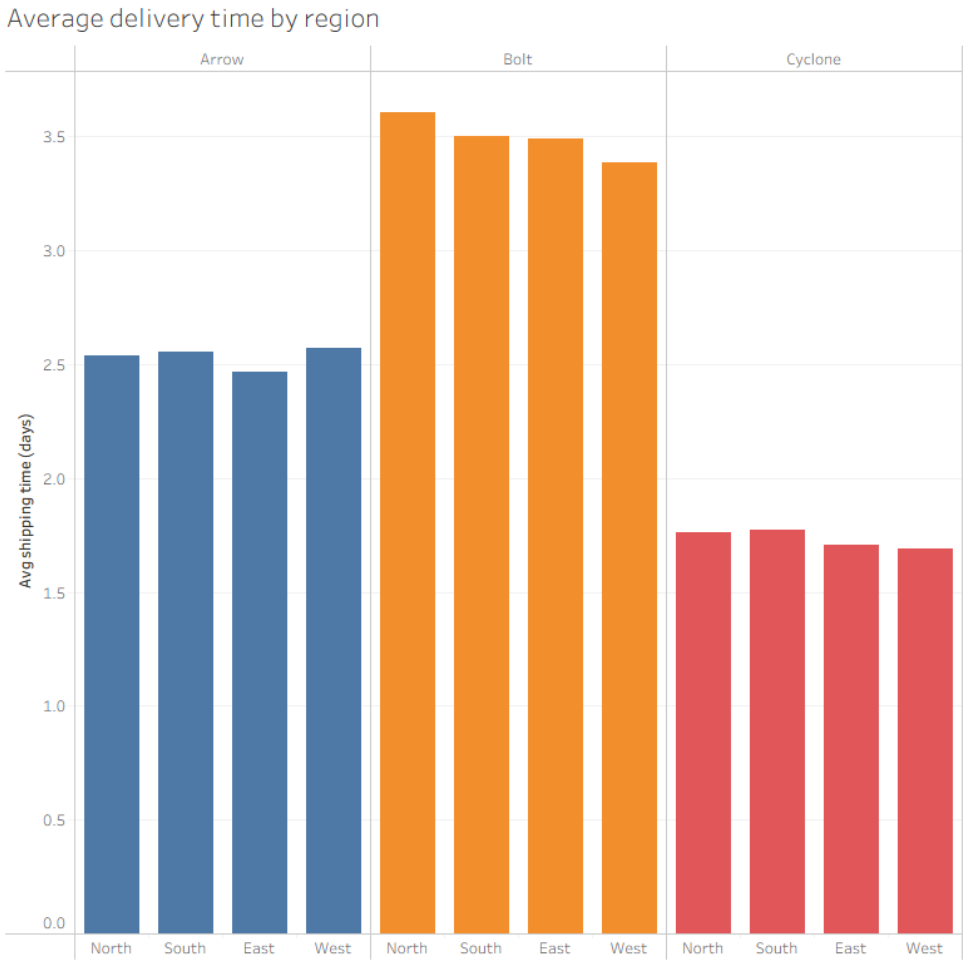
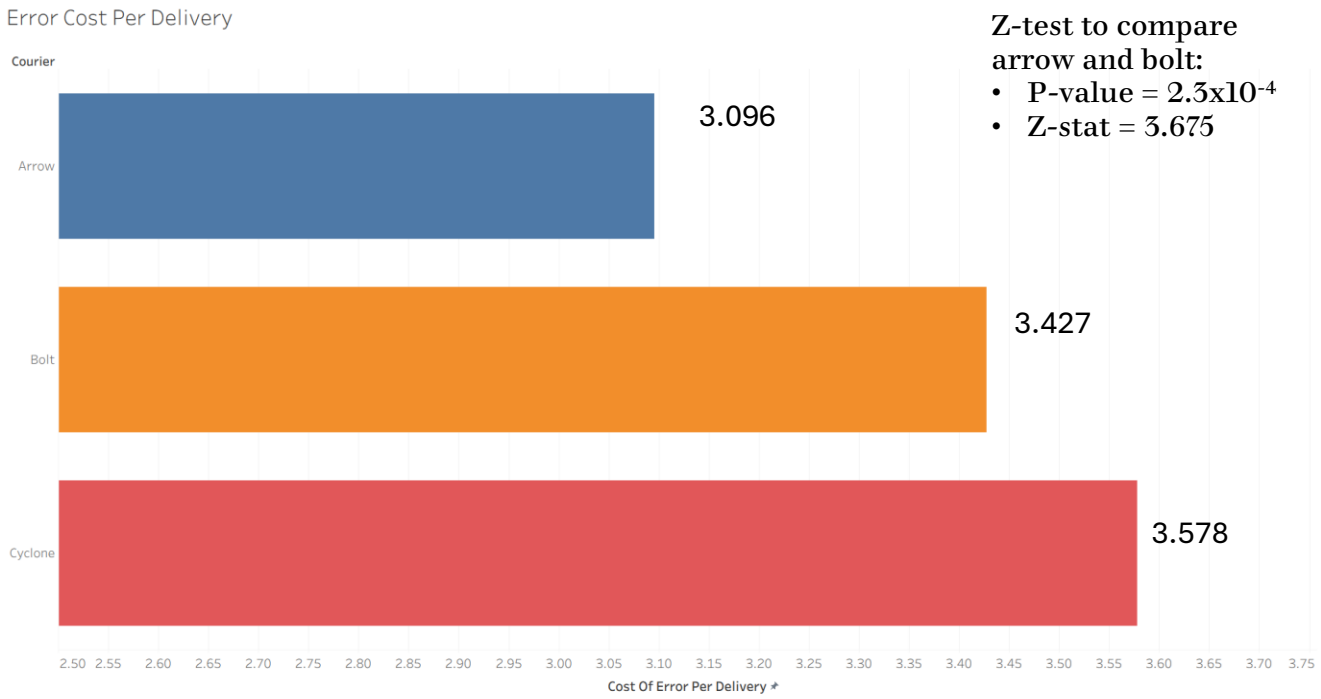
Oliver Darlington

Single Carrier Strategy - Arrow

If considering a single carrier to deploy across all 4 regions, the first consideration should be that the carrier is consistent and fast in their delivery times across all categories and more importantly regions.
Looking at the graph below, we can see all three couriers are consistent across the 4 regions. Analysis shows consistency across product categories in a similar pattern. Shown, Cyclone has the fastest average shipping time across all 4 regions.





Z-test to compare arrow and bolt:

- P-value = 2.3×10^{-4}
- Z-stat = 3.675

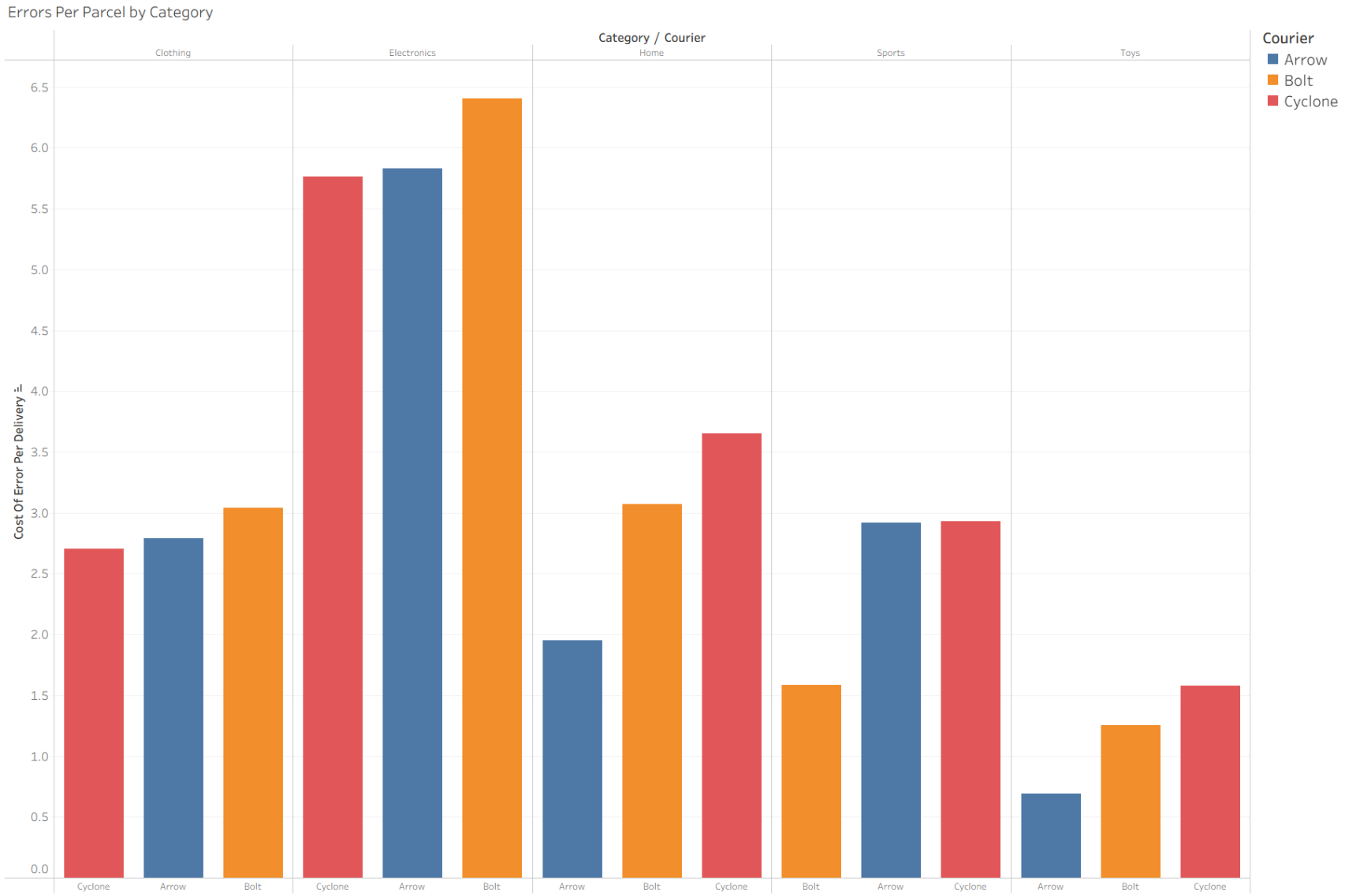
Being fast is only important if the courier is efficient in doing so. With the gap in parcels delivered between Bolt, delivering the most, and Arrow, the courier that delivers the least being only 47, I believe all of the carriers are viable options for a single carrier strategy. This led me to explore the number of parcels damaged, lost or late when delivered by an individual courier. We see the index of Arrow's loss per parcel being significantly lower than Bolt's the next most efficient firm.

This leads me to suggest if the firm were to take on a single carrier, it should be Arrow, the opportunity cost of day's faster delivery appears to be more than recuperated through the absence of courier error, however without knowledge of the amount charged by each courier for their service and the cost structure (daily rate, per parcel etc.) we have to assume the only varying factor between couriers is the losses in identified categories. Further exploration with an expanded dataset is necessary.

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Segmented Carrier Strategy – By Category

Knowing there is no specific edge for a single courier in each region, it leads us to segment courier by product category rather than region.



It appears from my exploration of the data that no one courier holds a large majority in each product category, leading me to believe we can assign based on minimal losses, avoiding Bolt where it statistically makes sense to do so, as to forgo their 2-day longer average delivery time when compared to Cyclone.

<u>Courier Assignment</u>

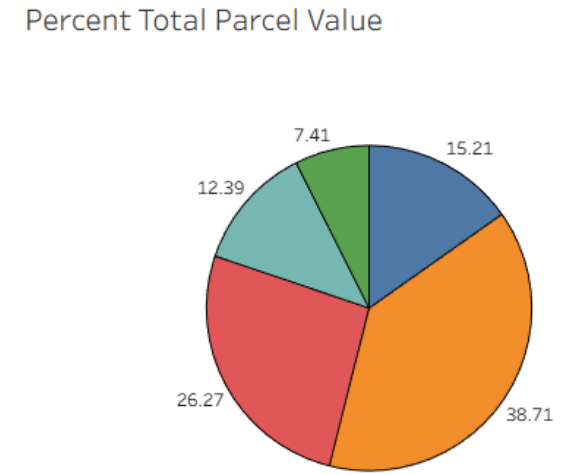
- Clothing - Cyclone
- Electronics - Cyclone
- Home - Arrow
- Sports – Bolt
- Toys – Arrow

The difference in loss per parcel in the sports category was statistically significant, with a z-test p-value of 7.794e-21, leading me to believe the operational savings are worth more than the 2-day slower delivery in comparison to cyclone.

Again, I assume all lost, damaged or late parcels are due to the fault of the courier and not the e-commerce seller.

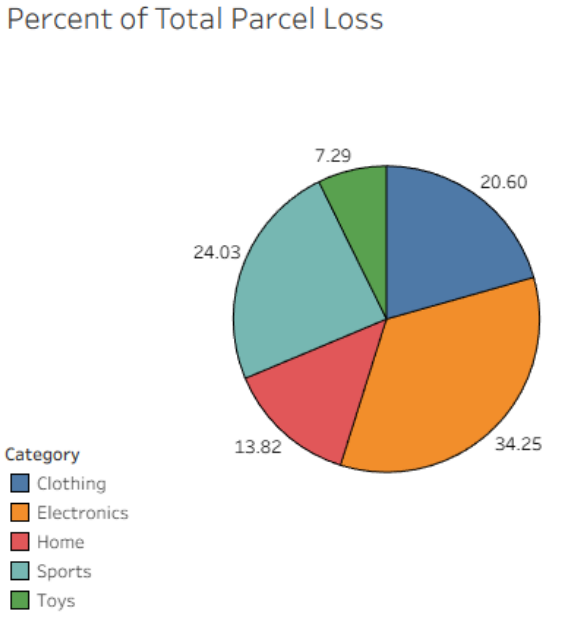
Oliver Darlington

Operational Improvements

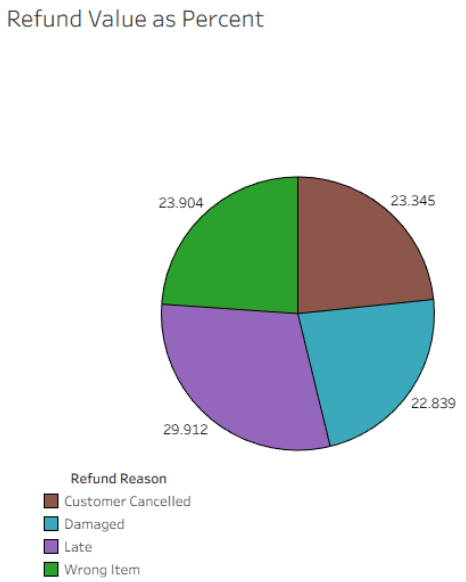


The first of my suggestions for an operational improvement looks to account for reducing the high percentage of total losses resulting from electronics.

For high ticket electronic items, I would like to consider an alternate method of customer collection such as partnering with a supermarket chain to offer click and collect. This will help to eliminate the 29% of electronics parcels that are returned due to damage as the parcels will be handled less and shipped in bulk. Furthermore, removing the ability of a customer to cancel an order will firstly make customers think more about an order before placing it, but also will eliminate just over 25% of the refund events and 23% of the refund value.



Acknowledging electronics is one of the more efficient categories in the fact it accounts for a smaller % of refund value than its total order value, a smaller saving from the category still accounts for a larger portion of the total return losses, meaning it is a prime target to yield higher operational savings.



My second suggestion for an immediate operational improvement is removing the ability of a customer to cancel an order before it is delivered. This will remove the burden of the transport and logistics costs of cancelled orders. While this fact may see a decrease in total order volume, the savings yielded by not unnecessarily moving parcels will more than account for this.

Cancellations account for just over 25% of the refund events and 23% of the refund value, this will eliminate a quarter unnecessary logistics overnight.

Comments on the Dataset

With the data at hand, it is very difficult to draw conclusions as to at what point in the supply chain errors occur. For example, is it a fulfilment error at fault of the courier or a packaging error, as the fault of the e-commerce firm if a customer receives the wrong item in a delivery. As a result of this, a substantial amount of assumptions must be made in order to inform courier strategies is staggering. Collecting data as to where an error occurred, although challenging will be incredibly useful.

The limitations of the dataset can lead the conclusions I have drawn from this dataset to be entirely different from that another individual may draw, and potentially incorrect to the real-world event. However, I believe the assumptions I have made and the factors I have chosen to examine reflect the best cases to deliver an analysis as close to realistic as possible despite the data limitations.